

USRSE 2026



Advancing Science in the Age of AI

San Jose, CA · October 19–21, 2026



Full program details and abstracts

<https://us-rse.org/usrse26/program/>

Welcome!

Placeholder. The General Chairs replace this file with their welcome letter before the program is printed, and remove `placeholder: true` above. Plain Markdown paragraphs, *emphasis*, **strong**, and links are supported; links print as their text.

Alex Koufos and Keith Beattie **General Chairs, USRSE'26**

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Schedule Overview

Full abstracts and presenter names are in the online program: <https://us-rse.org/usrse26/program/>

Monday, October 19

7:30–8:30am	Registration , <i>Prefunction</i>
8:30–10am	Welcome and Plenary Session , <i>San Jose Ballroom Salon 3 & 4</i> Keynote: Fernando Pérez
10–10:30am	Morning Break (Sponsorship available) , <i>Anywhere</i>
10:30–11:30am	Panel , <i>San Jose Ballroom Salon 3 & 4</i> Student & Early Career: Careers in RSE: Experiences from Research Software Engineers Across Industry, Academia, and Laboratories
10:30am–12pm	RAG Assistants , <i>San Jose Ballroom Salon 1 & 2</i> • AquiLLM: Evaluating Faithfulness in Open-Weight RAG-LLM Systems for Scientific Research • LZ Oracle: an AI-assisted knowledge retrieval platform for the LUX-ZEPLIN dark matter experiment • Data Sherpa: An AI-Powered Assistant for Scientific Collaboration Knowledge Management • AskALCF: A RAG Assistant for HPC User Support Case Studies Demonstrating RSE Impacts Part I , <i>San Jose Ballroom Salon 5 & 6</i> • Automating Data-Driven Alternative Text in Quarto-based NOAA Fisheries Reports • A Modular Threaded Software Application for Axion Detection with DMRadio • Poisonous Books: Rule Based Physics Pipeline and Agentic LLMs for Detecting and Analyzing Toxic Pigments in Victorian Bookbindings • Building Modern Tools for Computational Science in Flutter to Track the American Chestnut AI Assisted Code Development , <i>Willow Glen Room</i> • Reuse, Compose, Extend, Standardize, Automate: Two Decades of RSEing Open (Neuro)Science at CON • Research Software Engineering at SNS and HFIR: Advancing Neutron Science Software in the Age of AI-Assisted Development • Designing for Trust in AI-Assisted Scientific Computing
12–1:30pm	Lunch Break (Sponsorship available) , <i>San Jose Ballroom Salon 3 & 4</i>
1:30–3pm	RSE Team Operations , <i>San Jose Ballroom Salon 3 & 4</i> Birds of a Feather: Navigating RSE Team Operations: Leadership, Sustainability, and Emerging Challenges Daniel Madren, Geoffrey Lentner, Laura Theademan, Chris Bentley, Christina Maimone, Ian Cosden, Sandra Gesing, Adam Rubens, David Pettifor, Elizabeth Leake, William Cowen, Abbey Roelofs and Daniel Shown Institutional RSE Community , <i>San Jose Ballroom Salon 1 & 2</i> Birds of a Feather: Building an Institutional RSE Community Julia Damerow, J.C. Subida, Alex Koufos, Daniel Woulfin and Trevor Spreadbury Better Scientific Software Fellowship , <i>San Jose Ballroom Salon 5 & 6</i> Birds of a Feather: Better Scientific Software Fellowship Community Adam Lavelly, Elsa Gonsiorowski and Mary Ann Leung Threat Modeling for Research Software Systems , <i>Willow Glen Room</i> Workshop: A Group Attempt at Threat Modeling for Research Software Systems James Davis, George K. Thiruvathukal, Jeffrey Carver, Kelechi G Kalu and Alexandra Harris-Watson
3–3:30pm	Afternoon Break (Sponsorship available) , <i>Anywhere</i>

Monday, October 19, continued

3:30–5pm

Interactive Computing and Reproducible Workflows, *San Jose Ballroom Salon 1 & 2*

- Effective Tutorial Notebooks for Cloud Computing Workflows using Remote Sensing Datasets
- Marimo: an open-source reactive notebook for Python
- Beyond human readers: designing domain-specific language notebooks for AI-assisted code generation
- From Notebook to Control Room: Reproducible MLOps for Accelerator Digital Twins

Lessons from Training Users & RSEs, *San Jose Ballroom Salon 5 & 6*

- INTERSECT After Three Years: Lessons Learned from Delivering Advanced Research Software Engineering Bootcamps
- Beyond Workshops: RSE-Led Hackathons as Experiential Learning for Software Practice in the AI Era
- Building Open Source Communities Through Effective Tutorials
- Evolving HPC User Support: Just-in-Time Learning with Instructional Videos

AI + Software Quality, *Willow Glen Room*

- Software Sustainability Toolkit: Measuring and Improving Sustainable Research Software at Scale
- Measuring What Matters in the Age of AI: The Research Software Metrics Landscape
- Back to Basics: Design-First Development in the Age of AI Coding Assistants
- Scientific Open-Source Software Is Less Likely to Become Abandoned Than One Might Think: Lessons from Curating a Catalog of Maintained Scientific Software

5:30–7:30pm

Poster Reception, *San Jose Ballroom Salon 3 & 4*

Tuesday, October 20

7:30–8:30am

Registration, *Prefunction*

8:30–10am

Plenary II, *San Jose Ballroom Salon 3 & 4*

10–10:30am

Morning Break (Sponsorship available), *Anywhere*

10:30–11:30am

Making AI Sustainable: Token Economics, *San Jose Ballroom Salon 3 & 4*

- Agents on a Ramen Budget: Token-Smart Agentic Workflows with Local Models
- TBD

10:30am–12pm

RSE Practices in the Age of AI Part I, *San Jose Ballroom Salon 1 & 2*

- Research Software Engineering in the Age of Generative AI
- Beyond Editing Code: Uses for AI Tools in Scientific Work
- Ten Projects, Two Teams, One Question: How Does AI Impact Development?
- Coding is Easy, Everything else is Hard: Lessons learned from working with third-party software developers

The Human Factor, *San Jose Ballroom Salon 5 & 6*

- Usability Challenges across AI-Assisted Infrastructure, Security, and Scientific Workflows
- From Mood Boards to Web Components: Our Journey Building a Design System for Research
- Human in the Loop: Requirements Gathering and Researcher Trust in the Age of AI
- Designing for Two Audiences: AI-Readiness and Human Data Usability

HPC + AI, *Willow Glen Room*

- DART: Distributed Assignment of Research Tasks for Heterogeneous Compute Environments
- Ten Practices for Refactoring Bioinformatics Pipelines to nf-core DSL2 on Shared Slurm HPC
- Tapis AI Assistant: Increasing Research Impact through Traceable and Evaluated Agentic Tools
- Structuring agentic AI for HPC code modernization

Tuesday, October 20, continued

12–1:30pm	Lunch Break (Sponsorship available), <i>San Jose Ballroom Salon 3 & 4</i>
12:45–1:15pm	Working Group Fair (Presentations during lunch), <i>San Jose Ballroom Salon 3 & 4</i>
1:30–3pm	RSE Skill Training, <i>San Jose Ballroom Salon 3 & 4</i> Birds of a Feather: How to train an RSE: A discussion of RSE skill training - past, future, and present Ellianna Abrahams, Lauren Milechin, Jeffrey Carver, Daniel Woulfin, Patrick Diehl and Gabrielle O'Brien User Experience: Journey Mapping, <i>San Jose Ballroom Salon 1 & 2</i> Workshop: Journey Mapping: A technique to see opportunities and recognize gaps in user experience Johanna Cohoon, Rajshree Deshmukh, Md Hasan Shahriar Simanto, Mary Goldman, Katrina Turner, Sarah Poon Improve Research Outputs by Improving Institutional Collaboration, <i>San Jose Ballroom Salon 5 & 6</i> Birds of a Feather: How can institutional RSE hubs and OSPOs work together for the betterment of research outputs? William Gearty, Jeffrey Young, Collin Capano, Ian Cosden, Daniel S. Katz, Richard Littauer, Fang Liu and Daniel Shown Refactoring Code for Reuse and Impact, <i>Willow Glen Room</i> Workshop: Fixer-Upper Software: Refactoring Code for Reuse and Impact Brian Ward and Jeff Soules
3–3:30pm	Afternoon Break (Sponsorship available), <i>Anywhere</i>
3:30–5pm	Case Studies Demonstrating RSE Impacts Part II, <i>San Jose Ballroom Salon 1 & 2</i> • Josh: Efficient, Portable, and Scalable Cross-Disciplinary Vegetation Modeling via Domain-Specific Language • Take it on the road: How RCD Helps Field Researchers Report Culvert Condition even when Offline • Escaping the Enclave: Format-Faithful Synthetic Data for Open, Reproducible Medicare Claims Pipelines • Automated Data Reduction for Neutron Scattering Experiments: Current Capabilities and Future Directions Code Modernization Using AI, <i>San Jose Ballroom Salon 5 & 6</i> • Agentic code modernization • Scientific Agentic Engineering: A Framework for Reliable AI Agents in Research Software • From Laptop to Cluster: HPC Translation Patterns for AI Stacks • Beyond Code Pairs: Generating Test-Grounded Data for AI-Assisted Scientific Software Modernization Workflows, <i>Willow Glen Room</i> • Cloud-Native Graphical and Programmatic Cross-Database Access to Experiment Metadata with Trino, Superset and Ibis • Custom Containers for Continuous Integration via Spack: Decoupling Environment Configuration from Test Execution • OSC-IS: Research Software Infrastructure for Scientific Artifact Preservation, Provenance, and Reproducible Collaboration • Fabla: An Open-Source Voice-First EMA Platform for Clinical Research

Wednesday, October 21

7:30–8:30am	Registration , <i>Prefunction</i>
8:30–10am	Plenary III & Sponsor Talks , <i>San Jose Ballroom Salon 3 & 4</i> Announcements, US-RSE Updates, US-RSE Awards, BSSw Fellowship Awards
10–10:30am	Morning Break (Sponsorship available) , <i>Anywhere</i>
10:30am–12pm	Collaboration & Community Building , <i>San Jose Ballroom Salon 1 & 2</i> • Building Inclusive Talent Pathways: Scaling Outreach, Workforce Development, and Community Engagement for RSE • HPC Carpentry and Community Building at NIST • Navigating the Code Review Process as a Team: A Case Study • Collaboration within RSE group at Princeton: development of slurmise for resource usage estimation Software Quality , <i>San Jose Ballroom Salon 5 & 6</i> • Developing a Handbook of best practices for research software sustainability at Princeton • We Create Quality: Towards a Human-Centric Theory of Research Software Quality in the Age of AI • Beyond Productivity: Human-Centered Guardrails for AI-Assisted Research Software Workflows • Performance Portable Software Design Patterns – What they can do and why we should collect them. Generative AI + Applications , <i>Willow Glen Room</i> • From Chaos to Schema: LLM-Powered Document Information Extraction • Steering an LLM AutoResearch Loop with Domain Context: A Case Study with Vehicle Energy Models • Logic Meets Intuition: The Neuro-Symbolic Upgrade for LLMs
12–1:30pm	Lunch Break (Sponsorship available) , <i>San Jose Ballroom Salon 3 & 4</i>
1:30–3pm	RSE Practices in the Age of AI Part II , <i>San Jose Ballroom Salon 1 & 2</i> • AI-Augmented Research Software Engineering: Risks, Challenges, and Practices • A Framework for Evaluating AI-Integrated Research Software: Demonstration and Validation through CyberFacCES Smart Search • A Nonprofit's Google Summer of Code Experience in the Age of AI-Assisted Contributions • Preparing Students for AI-Assisted Research Workflows Reflections on RSE Roles & Pathways , <i>San Jose Ballroom Salon 5 & 6</i> • What They Don't Tell You About the Data Science to RSE Transition • Bears, Bottlenecks, and Bus Factors: Why Embedded RSEs Pay Off • Mission Control: Service and the Stewardship of Science Building Infrastructure for Communities or Organizations , <i>Willow Glen Room</i> • From Lakehouse to Labmate: KBase's Vision for AI-Enabled Scientific Discovery • HEROIC: A Research Software Infrastructure for Real-Time Multi-Messenger Observatory Coordination • A Managed Database-as-a-Service Platform for Scientific Computing • One Size Fits None: Rethinking Campus Research Computing
3–3:15pm	Transition Break , <i>Anywhere</i>
3:15–4:30pm	Rapid Access Microtalks (RAM), Closing Remarks , <i>San Jose Ballroom Salon 3 & 4</i>

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